

L7b: Utility-Based Portfolio Allocation

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you should be able to:

- Describe preferences with utility functions: explain how a utility function ranks alternatives and how a concave utility of wealth describes risk aversion. Derive the risk-free fraction that maximizes mean-variance utility, and infer a client's risk aversion.
- Solve budget-constrained utility problems: solve the Cobb–Douglas problem over share counts in closed form, and state how the CES elasticity moves the allocation.
- Compute preference weights from the single index model: combine intercepts and betas with two market inputs from SPY. Signs choose the basket, and magnitudes split the budget.

Lectures and Examples

Lecture: Utility-Based Portfolio Allocation

Example 1: Choose the Optimal Point on the Capital Allocation Line
Which risk-free fraction maximizes utility? Infer the risk aversion behind each interview answer.

Example 2: Build a Maximum-Utility Portfolio Allocator
Which firms belong in the basket, how much of each, and what happens when we re-solve the allocation daily and monthly through 2025 against buy-and-hold portfolios?

Both examples use the default firms. Uncomment the ticker-file lines to use the client's firms. Example 1 reads the client's risk-free fraction and income-gamble range when those files are present.

Concept Review: The Capital Allocation Line

A complete portfolio c holds w_f in the risk-free asset and $1 - w_f$ in the tangent portfolio $\mathcal{T}$:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_\mathcal{T}] - g_f) \qquad \text{expectation is linear}$$

$$\sigma^2_{g,c} = (1 - w_f)^2 \, \sigma^2_{g,\mathcal{T}} \qquad \text{the risk-free asset adds no variance}$$

Lending ($0 < w_f < 1$) and borrowing ($w_f < 0$) move c along the capital allocation line, whose slope is the Sharpe ratio of $\mathcal{T}$.

By two-fund separation, investors with the same inputs who can lend and borrow freely at g_f hold the same risky fund $\mathcal{T}$. Their risk preferences set only w_f, and the L6b client interview asked for w_f directly.

Which w_f should a given investor choose?

Company Profile: Wealthfront

Wealthfront is a robo-advisor that manages portfolios of low-cost index ETFs, with more than $100 billion in client assets in September 2026. Andy Rachleff and Dan Carroll founded it in 2008, and Burton Malkiel has been its CIO since 2012.

- Risk questionnaire: willingness and capacity to take risk combine into a Risk Score from 0.5 to 10, weighted toward the more risk-averse part.
- Utility and the frontier: twenty frontier portfolios, one per score, with expected returns from the Black–Litterman model (market-implied returns plus factor-model forecasts). In 2012, the firm described mapping the score to a risk-aversion coefficient and picking the portfolio where the indifference curve touches the frontier.

Today's connection: the questionnaire plays the role of the L6b client interview, and a higher Risk Score plays the role of a smaller A.

Explore: Careers Engineering intern program YouTube channel

Utility Functions and Rational Choice

Utility functions describe a decision-maker's preferences for different choices, outcomes, or states of the world. Utility is a proxy for satisfaction, so a larger utility is always better.

Utility function: an agent chooses from alternatives $\mathcal{X} = \{x_1, \dots, x_n\}$, e.g., bundles of goods, portfolios, or states of the world. $U : \mathcal{X} \to \mathbb{R}$ assigns each alternative a number, in *utils*, that ranks the agent's preferences:

$$U(x_i) > U(x_j) \iff x_i \succ x_j \qquad \text{and} \qquad U(x_i) = U(x_j) \iff x_i \sim x_j$$

- Ordinal: utility ranks alternatives but does not measure how much better one is. Any order-preserving transformation of U, e.g., $\ln U$ for positive U, describes the same preferences.
- Key idea: when faced with competing alternatives, a rational decision-maker always chooses the alternative with the largest utility.

Common Utility Functions

Utility functions, and their parameters, are subjective. Five common forms for a bundle of goods $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \geq 0$ and $\alpha_i > 0$:

Utility function	Form	Models
Linear	$U(\mathbf{x}) = \sum_i \alpha_i x_i$	goods that are perfect substitutes
Logarithmic ($\beta > 0$)	$U(\mathbf{x}) = \ln(\sum_i \alpha_i x_i + \beta)$	diminishing satisfaction from more goods
Cobb–Douglas	$U(\mathbf{x}) = \prod_i x_i^{\alpha_i}$	goods held together
Leontief	$U(\mathbf{x}) = \min(x_1/\alpha_1, \dots, x_n/\alpha_n)$	perfect complements, fixed proportions
CES ($\eta > 0$, $\eta \neq 1$)	$U(\mathbf{x}) = \left(\sum_i \alpha_i x_i^{(\eta-1)/\eta}\right)^{\eta/(\eta-1)}$	substitution set by the elasticity η

We use the Cobb–Douglas and CES utilities later today. As η changes, CES moves between the linear ($\eta \to \infty$), Cobb–Douglas ($\eta \to 1$, with $\sum_i \alpha_i = 1$), and Leontief ($\eta \to 0$) forms.

Risk and Risk Aversion

When outcomes are uncertain, a rational decision-maker ranks choices by the expected utility $\mathbb{E}[U(W)]$ (von Neumann–Morgenstern). Only $aU + b$ with $a > 0$ keeps this ranking, so the shape of the utility of wealth $U(W)$ sets the attitude toward risk:

Curvature of U	Risky versus certain	Risk attitude
Concave ($U'' < 0$)	$\mathbb{E}[U(W)] \leq U(\mathbb{E}(W))$	risk averse
Linear ($U'' = 0$)	$\mathbb{E}[U(W)] = U(\mathbb{E}(W))$	risk neutral
Convex ($U'' > 0$)	$\mathbb{E}[U(W)] \geq U(\mathbb{E}(W))$	risk seeking

The Certainty Equivalent

Picture a coin flip that leaves you with wealth W_1 or W_2, and ask two questions:

- What is the gamble worth on average? $\mathbb{E}(W)$, in USD. A risk-neutral decision-maker would trade the gamble for exactly this sure amount.
- What sure wealth W_3 would you take instead? The one that leaves you indifferent is the certainty equivalent, also in USD.

The CE is the sure wealth with the same utility as the risky outcome:

$$\underbrace{U(\mathrm{CE})}_{\text{utility of the sure wealth}} = \underbrace{\mathbb{E}[U(W)]}_{\text{expected utility of the risky outcome}} \qquad \text{definition of the CE}$$

$$\mathrm{CE} = U^{-1}(\mathbb{E}[U(W)]) \qquad U \text{ is increasing, so apply } U^{-1}$$

The Risk Premium

For a risk-averse decision-maker, the definition of the CE and concavity of U chain together:

$$\begin{aligned} U(\mathrm{CE}) &= \mathbb{E}[U(W)] && \text{definition of the CE} \\ &\leq U(\mathbb{E}(W)) && \text{concave } U \\ \mathrm{CE} &\leq \mathbb{E}(W) && U \text{ is increasing} \end{aligned}$$

The gap is the risk premium, the expected wealth given up to remove the risk:

$$\mathrm{RP} = \underbrace{\mathbb{E}(W)}_{\text{expected wealth}} - \underbrace{\mathrm{CE}}_{\text{sure wealth}}$$

The Certainty Equivalent: An Example

Let $U(W) = \ln W$, and flip a fair coin for a wealth of 50 or 150 USD:

$$\mathbb{E}(W) = \tfrac{1}{2}(50) + \tfrac{1}{2}(150) = 100$$ average wealth, USD

$$U(\mathbb{E}(W)) = \ln 100 \approx 4.6052$$ utility of the sure average, utils

$$\mathbb{E}[U(W)] = \tfrac{1}{2}\ln 50 + \tfrac{1}{2}\ln 150 \approx 4.4613$$ expected utility of the gamble, utils

$$\mathrm{CE} = U^{-1}(\mathbb{E}[U(W)]) = e^{4.4613} \approx 86.60$$ $U^{-1}(x) = e^x$ undoes $\ln$, USD

$$\mathrm{RP} = 100 - 86.60 = 13.40$$ risk premium, USD

The investor would accept 86.60 USD for certain in place of a flip that pays 100 USD on average. Logarithmic utility has diminishing marginal utility: the 50 USD gained adds less utility than the 50 USD lost takes away.

Arrow–Pratt Measures

How risk averse is a decision-maker? Arrow and Pratt (1964) measure the curvature of a utility of wealth with $U'(w) > 0$:

$$\underbrace{r(w) = -\frac{U''(w)}{U'(w)}}_{\text{absolute}} \qquad \underbrace{\bar{r}(w) = w\, r(w)}_{\text{relative}}$$

	$U(w)$	$r(w)$	$\bar{r}(w)$
Log	$\ln w$	$1/w$	1
Linear	$a\,w + b$	0	0
Power	w^τ/τ	$(1-\tau)/w$	$1-\tau$

$r > 0$: averse. $r = 0$: neutral. $r < 0$: seeking.

For a small risk, larger r means a lower CE:

$$\mathrm{RP} \approx \tfrac{1}{2}\, r(\mathbb{E}(W))\ \mathrm{Var}(W)$$

For a portfolio: with small returns, CE return $\approx$ expected return $- \frac{\bar{r}}{2} \times$ return variance. Mean-variance utility has this form, with A in the role of $\bar{r}$.

Mean-Variance Utility

We choose w_f by maximizing the investor's utility. Mean-variance utility rises with expected growth and falls with variance:

$$\boxed{U(w_f) = \mathbb{E}[g_c] - \frac{A}{2}\,\Delta t\,\sigma^2_{g,c}}$$

The risk-aversion coefficient $A > 0$ (dimensionless) sets how much the investor dislikes variance. As in L5a, $\Delta t\,\sigma^2_{g,c}$ is the squared volatility, in year^{-1}. On the line, both terms depend only on w_f:

$$U(w_f) = g_f + (1 - w_f)(\mathbb{E}[g_{\mathcal{T}}] - g_f) - \tfrac{A}{2}(1 - w_f)^2\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{substitute the line}$$

$$\frac{dU}{dw_f} = -\big(\mathbb{E}[g_{\mathcal{T}}] - g_f\big) + A\,(1 - w_f)\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{differentiate}$$

$$0 = -\big(\mathbb{E}[g_{\mathcal{T}}] - g_f\big) + A\,(1 - w_f^\star)\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{first-order condition}$$

The Optimal Complete Portfolio

Solving the first-order condition for the fraction in $\mathcal{T}$ gives:

$$\boxed{1 - w_f^\star = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{A\,\Delta t\,\sigma_{g,\mathcal{T}}^2}}$$

- A maximum: the second derivative, $-A\,\Delta t\,\sigma_{g,\mathcal{T}}^2$, is negative.
- What moves it: more excess growth $\mathbb{E}[g_{\mathcal{T}}] - g_f$ raises the fraction in $\mathcal{T}$. A larger A or $\sigma_{g,\mathcal{T}}^2$ lowers it, and doubling A halves it.
- Who lends and who borrows? The investor with $A_{\mathcal{T}} = (\mathbb{E}[g_{\mathcal{T}}] - g_f)/(\Delta t\,\sigma_{g,\mathcal{T}}^2)$ holds only $\mathcal{T}$. $A > A_{\mathcal{T}}$ lends at g_f. $A < A_{\mathcal{T}}$ borrows at g_f to buy more of $\mathcal{T}$.

Indifference Curves and Risk Aversion

Where does the optimum sit on the line? An indifference curve joins the portfolios with the same utility. Fixing U gives:

$$\mathbb{E}[g_c] = U + \frac{A}{2}\,\Delta t\,\sigma_{g,c}^2$$

Plotted against $\sigma_{g,c}$, each curve is a parabola that opens upward, and higher curves have higher utility. The optimal portfolio is where the highest curve the investor can reach touches the line.

The result also runs backward. If a client picks $w_f < 1$, the A that makes that choice optimal is:

$$\boxed{A = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{(1 - w_f)\,\Delta t\,\sigma_{g,\mathcal{T}}^2}}$$

Example 1: Choose the Optimal Point on the Capital Allocation Line

Utility over Holdings

Every investor so far holds the same risky fund. Instead, ask a consumer-choice question: given a budget and today's prices, which holdings does the investor prefer most?

The investor chooses share counts that maximize a holding utility, a preference over the shares held today at today's prices.

Unlike the mean-variance utility, a holding utility has no variance term. Risk enters through the preference weights, which depend on the SIM betas and the state of the market.

Cobb–Douglas Utility

The exponents are the preference weights $\gamma_i \in (-1, 1)$. The investor maximizes the product over every asset, subject to the budget and a small floor of $n_{\min} > 0$ shares on each holding:

$$\begin{aligned} &\underset{n_1,\dots,n_{|\mathcal{P}|}}{\text{maximize}} && U(n_1, \dots, n_{|\mathcal{P}|}) = \prod_{i \in \mathcal{P}} n_i^{\gamma_i} \\ &\text{subject to} && \sum_{i \in \mathcal{P}} n_i \, S_i(t) + \text{cash}(t) = W_{\mathcal{P}}(t) && \text{(budget constraint)} \\ & && n_i \geq n_{\min} \quad \forall \, i \in \mathcal{P}, \qquad \text{cash}(t) \geq 0 && \text{(share floor, no borrowing)} \end{aligned}$$

where n_i is the share count of asset i in the asset set $\mathcal{P}$, $S_i(t)$ its price (USD per share), $W_{\mathcal{P}}(t)$ the wealth, and $\text{cash}(t)$ the wealth not in shares (USD), which does not enter U. The floor keeps U finite, because $n_i^{\gamma_i} \to \infty$ as $n_i \to 0$ when $\gamma_i < 0$. The preferred set $\mathcal{A}^+ = \{i : \gamma_i > 0\}$ is the basket, and the rest form the non-preferred set $\mathcal{A}^- = \{i : \gamma_i \leq 0\}$.

Cobb–Douglas Utility: Floors and Net Budget

Only the ranking matters, so $\ln U$ has the same maximizer as U. With a non-empty basket, we set the floors and the cash first:

$$\frac{\partial \ln U}{\partial n_k} = \frac{\gamma_k}{n_k} \le 0 \qquad k \in \mathcal{A}^-, \text{ so more shares never raise } \ln U$$

$$n_k^\star = n_{\min} \qquad \text{spend every dollar above the floor on } \mathcal{A}^+$$

$$\text{cash}^\star(t) = 0 \qquad \text{a preferred share raises } \ln U, \text{ and cash does not}$$

Substituting both into the budget constraint leaves the net budget for the preferred set:

$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) + n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) = W_{\mathcal{P}}(t)$$

$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) \equiv W_{\text{adj}} > 0$$

A Lagrange multiplier λ then splits W_{adj} over the preferred set, assuming each preferred share count stays above its floor.

Cobb–Douglas Utility: Allocation

Adding the multiplier λ for the net budget gives:

$$\mathcal{L} = \textstyle\sum_{i\in\mathcal{A}^+} \gamma_i \ln n_i - \lambda\left(\sum_{i\in\mathcal{A}^+} n_i S_i(t) - W_{\text{adj}}\right)$$ floor terms are constant

$$0 = \gamma_i/n_i - \lambda\, S_i(t)$$ differentiate with respect to n_i

$$n_i S_i(t) = \gamma_i/\lambda$$ dollars in proportion to γ_i

$$W_{\text{adj}} = \textstyle\sum_{j\in\mathcal{A}^+} \gamma_j/\lambda$$ sum over the preferred set

$$1/\lambda = W_{\text{adj}} \Big/ \textstyle\sum_{j\in\mathcal{A}^+} \gamma_j$$ solve for the multiplier

Substituting $1/\lambda$ into $n_i S_i(t) = \gamma_i/\lambda$ gives:

$$n_i^\star = \underbrace{\frac{\gamma_i}{\sum_{j\in\mathcal{A}^+}\gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \qquad (i\in\mathcal{A}^+), \quad n_i^\star = n_{\min} \quad (i\in\mathcal{A}^-)$$

If a preferred count falls below $n_{\min}$, pin it at the floor, subtract its cost from W_{adj}, and solve again. If every preferred count stays above its floor, dollars are proportional to γ_i.

CES Utility: A Concentration Dial

CES adds one parameter to Cobb–Douglas, the elasticity $\eta > 0$. It controls how willingly the allocator trades one preferred asset for another. For $\eta \neq 1$:

$$U_{\mathrm{CES}} = \Big(\sum_{i \in \mathcal{A}^+} \gamma_i \, n_i^{(\eta-1)/\eta} \Big)^{\eta/(\eta-1)}$$

$$n_i^\star = W_{\mathrm{adj}} \, \frac{(\gamma_i / S_i(t))^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t) (\gamma_j / S_j(t))^\eta} \quad (i \in \mathcal{A}^+)$$

With the same floor check, share counts above the floors are proportional to the *bang for the buck* $\gamma_i / S_i(t)$ raised to η. Three limits:

- Unit elasticity ($\eta \to 1$): the Cobb–Douglas allocation.
- Large elasticity ($\eta \to \infty$): the asset with the largest $\gamma_i / S_i(t)$, when unique, gets the whole budget above the floors.
- Small elasticity ($\eta \to 0^+$): equal share counts of every preferred asset.

Proofs: CES companion, which also shows that CES dollar weights depend on share price levels when $\eta \neq 1$.

Preference Weights from the Single Index Model

Each trading day, from the L6b SIM parameters (α_i, β_i), the recent market growth rate $\tilde{g}_{M,t}$, and the market-state signal ξ_t:

$$\gamma_i(t) = \tanh\left(h \left[\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t}\, \tilde{g}_{M,t} \right] \right)$$

The argument is $h\,\tilde{g}_{i,t}/\beta_i^{\xi_t}$, where $\tilde{g}_{i,t} = \alpha_i + \beta_i \tilde{g}_{M,t}$ is the SIM's expected growth rate of asset i today. The horizon $h = 1$ year makes it dimensionless, and the $\tanh$ keeps the weight in $(-1, 1)$. The model assumes $\beta_i > 0$.

- Sign, which assets are in the basket: $\gamma_i(t) > 0$ exactly when $\tilde{g}_{M,t} > -\alpha_i/\beta_i$, the threshold. As $\tilde{g}_{M,t}$ rises, the basket grows. As it falls, only the lowest-threshold assets stay. If the basket is empty, hold the floors plus cash.
- Magnitude, how much each asset gets: the magnitude sets the budget share. Bearish ($\xi_t > 0$) tilts it toward low-beta assets, bullish toward high-beta.

Exponential Moving Averages

Both market inputs come from the SPY price through exponential moving averages (EMAs). An EMA is a running average that gives recent prices the most weight. For a price series S_t and a window of L trading days:

$$\bar{S}_t = \omega\, S_t + (1-\omega)\, \bar{S}_{t-1} \qquad \text{where} \qquad \omega = \frac{2}{L+1}$$

This is the mean update of L5a with $\omega = 1 - \lambda$, but here a window L sets ω instead of a half-life. A 21-day window has a half-life of about 7 days, shorter than the 21-day half-life used in L5a.

The Market-State Signal

With $L_{\text{short}} = 21$ and $L_{\text{long}} = 63$ days and a gain $G > 0$, the market-state signal is the scaled crossover of the two price EMAs:

$$\boxed{\xi_t = -G\left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1\right)}$$

The short EMA follows the last few weeks and the long EMA the last quarter, so the ratio says where recent prices sit against the quarter's average:

- $\bar{S}_t^{\text{short}} > \bar{S}_t^{\text{long}}$: $\xi_t < 0$, bullish. Recent prices sit above the quarter's average: the market has been rising.
- $\bar{S}_t^{\text{short}} = \bar{S}_t^{\text{long}}$: $\xi_t = 0$, neutral, no tilt. A flat market, or the moment a trend reverses and the two EMAs cross.
- $\bar{S}_t^{\text{short}} < \bar{S}_t^{\text{long}}$: $\xi_t > 0$, bearish. Recent prices sit below the quarter's average: the market has been falling.

The Recent Market Growth

We choose G on the training data so that $|\xi_t| \leq 1$ on 99% of training days. The second market input is the recent market growth rate $\tilde{g}_{M,t}$:

- $\tilde{g}_{M,t}$ is the EMA ($L_{\text{growth}} = 252$ days, half-life about 87) of the daily market growth rates $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$ (inverse years).
- With a short window the estimate $\tilde{g}_{M,t}$ swings by over one per year and can empty the basket. A one-year window moves it by tenths.
- Both inputs on day t use prices through day $t-1$, so a decision never uses a price it could not have known.

Example 2: Build a Maximum-Utility Portfolio Allocator

Optional Advanced Material

The advanced index describes the optional notebook:

- CES limits and log-linear utility: derive the closed-form CES optimum and prove its three limits. Show that log-linear utility has the Cobb–Douglas optimizer, and why CES weights depend on share price levels.

Summary

- Risk aversion picks the point on the capital allocation line: we measured risk aversion with the certainty equivalent and the Arrow–Pratt coefficient and found the utility-maximizing risky fraction in closed form. More risk-averse investors lend more, and a client's chosen fraction implies the coefficient that income-gamble answers bracket.
- Cobb–Douglas turns preferences into weights in closed form: the preferred budget is split in proportion to the positive preference weights. The CES elasticity moves the allocation between concentration and equal share counts.
- The single index model and the market set the preferences: the recent market growth decides which assets belong in the basket, and the crossover signal tilts the budget between high-beta and low-beta assets.

Risk aversion picks the point on the capital allocation line, and preference weights pick the assets and their weights.